MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, FUEL, 0-1000 PSI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating 320-degree movement, synchro style, differential, fuel pressure gage transmitter.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
MIL-I-7088	Indicator, Pressure, Fuel, 0-1000 PSI
MIL-T-5350	Transmitters, Synchro, Operated Aircraft, General Specification For
MIL-P-26514	Polyurethane Foam, Rigid Or Flexible, For Packaging

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-2073-1	DOD Material Procedures For Development And Application Of Packing Requirements
MS28005	Transmitters, Pressure, Synchro, Aircraft
MS28010	Indicators, Pressure-Synchro, Single, 2-Inch Size

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620
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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society For Testing And Materials (ASTM)

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

3. REQUIREMENTS

3.1 First article. When specified, in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.5 and 6.3).

3.2 General. The requirements specified in MIL-T-5350 are applicable as requirements of this specification. Additional requirements shall be as specified herein. Where the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.3 Design and construction. The fuel pressure transmitter shall be designed for use with MS28010-4 pressure indicators conforming to Specification MIL-I-7088.

3.3.1 Hermetically sealed case. The case shall provide a hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing the case. This shall be accomplished without the use of special tools and fixtures unless they are approved by the contracting activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.3.2 Operation. Each transmitter shall be capable of operating in either a horizontal or vertical position.

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3.4 Dimensions. The outline dimensions of the transmitter shall conform to MS28005.

3.5. Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.6 Identification of product.

3.6.1 Nameplate. The nameplate shall be marked with the following information:

TRANSMITTER, PRESSURE, FUEL, 0-1000 PSI
26V Single Phase, 400 Cycles
MS Part No
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contractor Order No.
Manufacturer's Name or Trade-Mark
US Property

3.7 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 General. Methods of sampling, inspection and tests shall be in accordance with MIL-T-5350 and as specified herein.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5)

b. Quality conformance inspections (see 4.6)

4.4 Inspection conditions. Unless otherwise specified, the transmitter shall be inspected in a horizontal position with the vent connection at the lowest point.

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4.4.1 Tolerances. When the transmitter is subjected to the tests specified in MIL-T-5350, the tolerance shall not exceed those specified in tables I and II.

4.5 First article inspection.

4.5.1 Test samples. The test samples shall consist of five transmitters representative of the production equipment. The transmitters shall be subjected to all the tests in 4.6 of this specification and 4.5 of the general specification MIL-T-5350.

4.5 2 First article tests. The first article tests shall consist of all tests under 4.6 of this document and 4.5 of the general specification MIL-T-5350.

4.6 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests. Individual tests shall consist off all tests listed under 4.3.1 of MIL-T-5350.

b. Sampling tests. Sampling tests shall consist of all tests listed under 4.3.2 of MIL-T-5350 and 4.7 of this specification.

4.7 Sampling tests

4.7.1 Vibration. The amplitude of pointer oscillation of the master indicator shall not exceed the tolerances specified in table II. Scale errors shall not exceed the tolerances specified in table I.

4.7.2 Differential pressure error. The transmitter shall be tested for scale error with a pressure of 150 psi applied to the vent connection of the transmitter. The test shall be accomplished by subjecting the transmitter to the pressure specified to produce the indicator readings shown in table III. The scale errors shall not exceed the tolerances specified in table III.

4.7.3 Overpressure. The transmitter shall be subjected to 1,200 psi pressure for a period of 10 minutes. Not less than 1 hour following the application of the overpressure, the transmitter shall be tested for scale errors on increasing pressures only. The scale errors shall not exceed the tolerances specified in table I.

4.8 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

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TABLE I. Fuel pressure transmitter tolerances.

Test Pressure psi	Room Temperature Tolerances	Humidity Salt Spray Vibration	Overpressure Low Temperature High Temperature Seasoning, Endurance Altitude
0	5	15	15
50	5	10	10
100	8	15	15
200	10	20	20
300	10	20	20
400	15	25	30
500	15	25	30
600	15	25	30
700	15	25	30
800	15	30	30
900	20	30	40
1,000	20	30	40

TABLE II. Fuel pressure transmitter.

Test	Tolerance psi
Friction:	
Room Temperature	10.0
High Temperature	10.0
Low Temperature	25.0
Salt Spray	10.0
Humidity	20.0
Position Error at 500 psi	10.0
Vibration:	
Pointer Oscillation	20.0

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TABLE III. Differential pressure error.

Vent Connection	Test Pressure psi Pressure Connection	Gage Indication	Tolerance psi
150	150	0	20
150	250	100	20
150	350	200	20
150	450	300	20
150	550	400	25
150	650	500	25
150	750	600	25
150	850	700	25
150	950	800	25
150	1,050	900	25
150	1,150	1,000	30

5. PACKAGING

5.1 Preservation. Preservation shall be level A, C, or Industrial, IAW MIL-STD-2073-1 as specified (See 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning. Transmitters shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitters shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Preservatives shall not be used.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Each transmitter shall be placed in a PPP-B-636 Fiberboard container weather resistant, with sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

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5.1.3 Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size minimum tare and cube consistent with the protection required.

5.2.2 Level B. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affordds adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fuel pressure transmitter covered by this specification is intended for use to transmit the indicated pressure in the fuel system of aircraft engines to an applicable indicator.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. The quantity, part number and class number of the instrument desired.

c. Whether sampling plan B tests are to be conducted.

d. Reinspection date marking.

e. Level of packaging (5.1) and packing (5.2) desired.

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6.3 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

6.4 Subject term (key word) listing.

Fuel
MS28010-4
Pressure
Synchro
Transmitter
0-1000 PSI

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force -99
Army -AV

Preparing activity:
Air Force -71

Project number:
6620-0435

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-5396C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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